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RESEARCH PROFILE

M.Tech graduate in Computer Science Engineering at Calcutta University with research interests in Natural Language Processing, Digital Forensics, and Machine Learning. I have a strong passion for research and innovation, having explored diverse areas of Machine Learning and am now focused on advancing my expertise in Computer Vision and Natural Language Processing. Driven by curiosity and a commitment to impactful research, I aim to contribute to cutting-edge developments in AI and technology. I am interested in developing principled AI systems by combining data-driven learning with insights from mathematics, physics, and domain knowledge to improve robustness, interpretability, and reliability. My research experience spans NLP and computer vision, including language understanding and retrieval-based fact verification, as well as physics-based image forensics and detection of synthetic and manipulated media. My long-term goal is to advance trustworthy, knowledge-driven AI systems that can learn, reason, and generalize reliably across challenging scientific and real-world problems.

EDUCATION

University of Calcutta

Kolkata, India

Master of Technology (M.Tech.) in Computer Science & Engineering

Aug 2024 – Jun 2026

CGPA: 8.53 / 10.0 (First Class with Distinction) | **Academic Standing:** Among the top 7% of the cohort.

Dissertation: PHORENSICS: Physics-Based Framework for Deepfake Detection (10/10)

Relevant Coursework: Advanced Algorithms, Machine Learning, Image Processing, Advanced Database Systems, Distributed Systems, Operations Research

University of Calcutta

Kolkata, India

Master of Science (M.Sc.) in Computer Science

Dec 2022 – Jun 2024

CGPA: 7.48 / 10.0

Thesis: BEN-RS-ANN: An Innovative Approach for Revealing Emotion from Bengali Text (10/10, Grade 'O' – Outstanding)

Relevant Coursework: Artificial Intelligence, Image Processing & Pattern Recognition, Digital Signal Processing, Distributed & Cloud Computing, Advanced Database Management Systems

The Bhawanipur Education Society College

Kolkata, India

Bachelor of Science (B.Sc.) (Honours) in Computer Science

May 2019 – Jun 2022

CGPA: 8.59 / 10.0 (First Class with Distinction, A+)

Capstone Project: Accommodation Rent Prediction in India (A+ Grade)

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems

HONORS & ACHIEVEMENTS

Academic Excellence Award | *University of Calcutta & BESC*

2019 – 2026

- Graduated with First Class distinction in M.Tech., with academic performance placing among the top 7% of the cohort.
- Secured perfect 10/10 Grade Points across critical research-intensive modules, including M.Tech. Dissertation I & II, M.Sc. Project, and M.Sc. Artificial Intelligence Practicals.

Certificate of Appreciation | *Scholarly Journal Publication & Academic Achievement*

Silver Medal - Class Topper (Rank 2) | *SOF National Cyber Olympiad*

International Mathematics Olympiad (IMO) | *Science Olympiad Foundation*

Feb 2018

- Achieved International Rank 726 and Zonal Rank 31 (West Bengal & Andaman and Nicobar Zone) in the Science Olympiad Foundation International Mathematics Olympiad.
- Awarded a Certificate of Merit for outstanding performance in mathematics.

PEER-REVIEWED PUBLICATIONS

1. Seal, T., **Bhattacharya, A.**, Das, S., Patra, S., Dawn, D. D., Khan, A., & Setua, S. K. (2025). "BEN-RS-ANN: An Innovative Approach for Revealing Emotion from Bengali Text with Exposure to Polysemy Resolution." *Applied Computing for Software and Smart Systems (Proceedings of ACSS 2024)*. Springer LNNS, pp. 207-223. [**Scopus Indexed**]
 - Co-authored the proposal of a novel Resource-Scarce Artificial Neural Network (RS-ANN) architecture, contributing directly to the custom hidden layer selection method and the mathematical resolution of polysemous word ambiguity.
2. **Bhattacharya, A.**, Singhanian, A. V., Banerjee, P., Majumdar, R., & Bhounik, D. (2023). "A User Independent Recommendation System for Web Series." *Emerging Technologies in Data Mining and Information Security (Proceedings of IEMIS 2022)*. Springer LNNS. pp. 595-603. [**Scopus Indexed**]
 - Co-developed a personalized recommendation engine that localizes similarity clustering strictly within user-defined parameters, successfully mitigating the influence of external dataset bias.
3. **Bhattacharya, A.**, Dutta, S., & Bandyopadhyay, S. K. (2020). "Stock Market Price Prediction using two-layer LSTM-RNN." *International Journal of Advances in Engineering and Management (IJAEM)*, 2(1), 84-88.
 - Co-authored the implementation of a two-layer Long Short-Term Memory (LSTM) Recurrent Neural Network designed to capture time-varying, dynamic dependencies in sequential financial data.
 - Extracted and normalized raw market metrics to train a predictive model via an RMSprop optimizer, successfully forecasting the closing, opening, high, and low stock prices for Tesla, Inc. with a peak prediction accuracy of 89.82%.
4. **Bhattacharya, A.**, Singhanian, A. V., Banerjee, P., & Halder, S. (2020). "Statistical Inference of the Effectiveness of Lockdown for COVID 2019 in India." *International Research Journal of Engineering and Technology (IRJET)*, 7(5), 5920-5925.

- Utilized continuous curve-fitting approximations of the SIR epidemiological model to evaluate COVID-19 infection growth rates prior to and during lockdowns across ten Indian states.
- Calculated the relative change in infection growth rates using SciPy optimization routines, mathematically quantifying lockdown effectiveness independent of population parameters.

SELECTED RESEARCH & OPEN-SOURCE PROJECTS

PHORENSICS: Physics-Based Framework for Image Forgery Detection *Python, PyTorch, Docker, PySpark, PyPI*
Computer Vision, Trustworthy AI | Aug 2025 – Jun 2026

- Developed an open-source Python framework (phorensics) for physics-based image forgery detection, designed to provide interpretable alternatives to purely data-driven approaches.
- Integrated frequency-domain analysis, Error Level Analysis (ELA), and sensor pattern noise (PRNU) to identify artifacts introduced by generative image synthesis.
- Designed a fault-tolerant execution pipeline that dynamically switches between GPU and CPU computation, ensuring reliable execution under memory constraints.
- Built a distributed processing framework using Docker and PySpark to enable scalable forensic analysis across large image datasets.

AI-Powered Fact Verification System *Python, Gemma 3, SentenceTransformers, FAISS*
Natural Language Processing | Oct 2025

- Developed a retrieval-augmented fact verification framework combining LLM reasoning with semantic document retrieval for evidence-based claim verification.
- Proposed an adaptive convergence strategy that terminates iterative retrieval once semantic similarity stabilizes, reducing unnecessary inference while preserving prediction quality.
- Improved the FEVER benchmark by 8% Precision and 10% F1-score through cross-document evidence aggregation and optimized semantic similarity scoring.

TxtVis: Multimodal Representation Learning and Visualization *PyTorch, Streamlit*
Representation Learning

- Developed a from-scratch PyTorch framework for learning and visualizing latent representations across NLP and Computer Vision.
- Implemented Skip-Gram embeddings and a Siamese Convolutional Neural Network trained with Contrastive Loss to investigate representation geometry in multimodal embedding spaces.
- Built an interactive visualization interface for monitoring PCA projections, embedding evolution, and training dynamics during optimization.

Autonomous Vehicle Navigation System *Python, OpenCV, Flask, Socket.IO*
Deep Learning for Autonomous Driving | Jul 2020 – Jan 2021

- Developed an end-to-end autonomous driving system inspired by the NVIDIA DAVE-2 behavioral cloning architecture for steering angle prediction.

- Designed image preprocessing and augmentation pipelines, including YUV color transformation and randomized affine augmentation, to improve model generalization.
- Implemented a real-time telemetry framework for streaming vehicle data and evaluating closed-loop steering predictions.

RESEARCH EXPERIENCE

University of Calcutta

Research Intern (Media Forensics) | Under Prof. Sanjit Kumar Setua

Kolkata, India

Aug 2025 – Feb 2026

- Worked on a research project in Media Forensics, focusing on the analysis and verification of digital media to identify potential manipulation, tampering, and authenticity-related issues.
- Contributed to research involving image/video analysis, computer vision techniques, and machine learning-based approaches for detecting manipulated media.

University of Calcutta (Computer Science Department)

Research Intern (Academic) | Under Prof. Rajat Kumar Pal

Kolkata, India

Sep 2023 – Jul 2024

- Developed a Bengali NLP project for emotion detection with polysemy resolution, addressing the challenge of identifying the intended meaning and emotional context of words with multiple senses in a resource-scarce Bengali language environment.
- The project involved dataset creation, contextual feature selection, TF-IDF feature extraction, and development of the BEN-RS-ANN (Resource-Scarce Artificial Neural Network) model, designed to work efficiently with limited linguistic resources.
- Evaluated using 10-fold cross-validation, the proposed approach achieved 90.08% accuracy, outperforming the other evaluated classifiers.

The Bhawanipur Education Society College

Research Collaborator

Kolkata, India

Sep 2021 – Jan 2022

- Collaborated on interdisciplinary research between The Bhawanipur Education Society College and the Indian Statistical Institute (ISI), Kolkata, on a machine-learning-based user-independent recommendation system for web series.
- Developed a recommendation approach capable of generating personalized web-series recommendations using user-defined and localized parameters without requiring extensive user-history data.
- Applied similarity, correlation, and Jaccard-index-based techniques to address the challenge of producing personalized recommendations while reducing dependence on other users' preferences.

The Bhawanipur Education Society College

Undergraduate Research Intern

Kolkata, India

May 2020 – Sep 2020

- Conducted research and development in Machine Learning, Computer Vision, and COVID-19 data analytics under faculty mentorship.
- Worked on multiple COVID-19-focused projects, including a polynomial regression-based model for forecasting gender-wise COVID-19 infection trends using population-density data, statistical analysis to quantify the effectiveness of India's lockdown across different states, and a computer-vision-based face-mask detection project aimed at identifying mask compliance during the pandemic.
- Applied data preprocessing, mathematical/statistical modeling, regression and curve fitting, predictive analysis, and computer vision techniques to address real-world pandemic-related challenges.

ACADEMIC EXPERIENCE

Youth Career Hub

Project Instructor

Kolkata, India

Jul 2025 – Jul 2026

- Engaged in guiding and co-developing Computer Vision projects, fostering collaborative learning and strengthening applied skills alongside academic research.
- Spearheaded the design of scalable machine learning pipelines, mentoring students through the lifecycle of transitioning theoretical academic research into robust code implementation.
- Directed rigorous technical code reviews, explicitly advocating for the separation of core Python logic from external Deep Learning dependencies (PyTorch, Keras) to ensure long-term maintainability of image processing systems.

University of Calcutta

Teaching Assistant

Kolkata, India

Jul 2024 – Dec 2024

- Conducted Python programming laboratory sessions for undergraduate B.Tech. students.
- Guided students in implementing programming concepts, debugging code, and completing laboratory assignments.
- Demonstrated practical applications of Python, including object-oriented programming, file handling, and fundamental data structures.

TECHNICAL SKILLS

Core Programming: Python, C++, SQL, Object-Oriented Programming (OOP), Algorithm Design, Python Library Packaging (PyPI), Hardware/Memory Management

Libraries & Frameworks: PyTorch, OpenCV, Apache Spark, Keras/TensorFlow, Streamlit, CUDA

DevOps & Architecture: Docker, Git, Linux/UNIX Environments, PySpark Cluster Distribution, RESTful APIs (Flask)

Top Skills: Recommender Systems, Data Mining, Statistical Data Analysis

Languages: English (Full Professional), Bengali (Native or Bilingual), Hindi (Professional Working)

Research Interests: Natural Language Processing • Computer Vision • Machine Learning • Trustworthy AI • Human-Centered AI • AI for Science